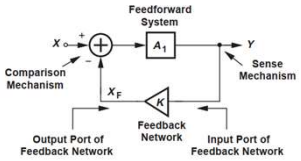
Feedback

- Feedback is a fundamental concept in electronics and many other fields.
- It involves taking a fraction of the output signal of a system and feeding it back to the input.
- This process allows for the regulation and precise control of a system's behavior.
- In electronics, feedback is used to design stable and predictable circuits, especially amplifiers.
- For instance, an amplifier with a very high but unstable gain can be made into a precise and stable amplifier using negative feedback.

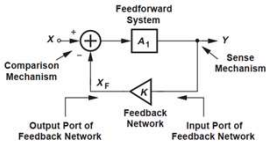
General Concept of Feedback

1. **Feedforward System (A_1):** This is the main, often high-gain and poorly controlled, part of the system. In an electronic circuit, this is typically an amplifier.
2. **Sense Mechanism:** This component measures the output signal (Y).



1

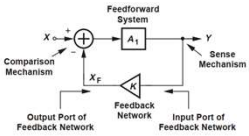
3. **Feedback Network (K):** This network processes the sensed output to create a feedback signal X_F.
   •The ratio $K = X_F / Y$ is known as the "feedback factor."
4. **Comparison Mechanism:** Here we subtract the feedback signal (X_F) from the input signal (X) to produce an "error" signal:
   • $E = X - X_F$ which then becomes the input to the feedforward system.



- **Important**
- The feedback is termed "negative" because the feedback signal is subtracted from the input.
- This corrective action is the basis for the stability and precision offered by negative feedback.

2

- The overall gain of this system, known as the **closed-loop transfer function**, can be derived as follows:
  - WKT, Transfer function (TF) =Y/X
  - And $X_F = KY$
  - And $E = X - X_F = X - KY$
  - This serves as the input of the forward system
  - ➔ $(X - KY)A1 = Y$
- Now move all the terms with Y to the LHS, and keep X RHS:
  - $Y(1+A1K) = A1X$



$$\frac{Y}{X} = \frac{A_1}{1 + KA_1}$$

- This fundamental equation reveals that negative feedback reduces the overall gain from A_1 ➔ $A_1 / (1 + KA_1)$.
- The quantity $\mathbf{KA_1}$ is called the **loop gain** and is a critical parameter in feedback analysis.

3

Types of Feedback

•Negative Feedback: Opposes input, provides stability
•Positive Feedback: Reinforces input, used in oscillators

Properties and Benefits of Negative Feedback

1. Gain Desensitization
2. Bandwidth Extension
3. Modification of I/O Impedances
4. Linearity Improvement

1. Gain Desensitization

Problem with open-loop gain system :

- Consider A CS amp.
- WKT *its* $Av = g_m * R_D$.
- Now both gm and RD are vary with process and temperature ➔ gain may vary as much as ±20% .
- So precise control of gain is not possible here.

4

The Solution: Negative feedback makes the closed-loop gain relatively independent of the open-loop gain, A_1.

That is wkt

$$\frac{Y}{X} = \frac{A_1}{1 + KA_1}$$

Now, if $KA1 \gg 1$ ➔ $(1 + KA_1) \approx KA_1$

So we have $\frac{Y}{X} \approx \frac{1}{K}$

- The above example reveals that the closed-loop gain of a feedback circuit becomes relatively independent of the open-loop gain so long as the loop gain, $KA1 \gg 1$.
- This property of negative feedback is called "**gain desensitization**".
- The feedback network is typically built with passive components like resistors, whose ratios can be controlled with high precision.

5

Why are resistor ratios more precise?

- Suppose if resistors R_1 and R_2 are i/p and feedback resistors (or vice versa) are used, and are made of the same material and fabricated close together on an integrated circuit, then they will "track" each other.
- This means any variations due to temperature or manufacturing will affect both resistors proportionally, leaving their ratio, R_1/R_2 or R_2/R_1, almost constant.

6

Gain Desensitization

- **Problem:** The circuit shown is designed for a nominal gain of 4.
  - (a) Determine the actual gain if $A_1 = 1000$.
  - (b) Determine the percentage change in the gain if A_1 drops to 500.

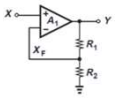
Solution:

- Nominal gain

$$\frac{Y}{X} \approx \frac{1}{K} \approx 1 + \frac{R_1}{R_2} = 4$$

  implies that $\boldsymbol{R_1/R_2 = 3}$.
- (a) The actual gain is given by

$$\frac{Y}{X} = \frac{A_1}{1 + KA_1} = \mathbf{3.984}$$



- Note that the loop gain $KA_1 = 1000/4 = 250$.
- (b) If A_1 falls to 500, then

$$\frac{Y}{X} = \frac{A_1}{1 + KA_1} = \mathbf{3.968}$$

- Thus, if A_1 drops by factor of 2
  - The closed-loop gain changes by only

(3.984 - 3.968)/3.984 = **0.4%**

7

2. Bandwidth Extension

- Every amplifier has a limited frequency range over which it operates effectively.
- This range is called its bandwidth.
- For a simple amplifier, the gain starts to "roll off" at a certain cutoff frequency.

Let us consider a one-pole open-loop amplifier with a transfer function

Substitute $A_1(s) = \dfrac{A_0}{1 + \dfrac{s}{\omega_0}}$ into $\dfrac{Y}{X} = \dfrac{A_1}{1 + KA_1}$

$$\frac{Y}{X} = \frac{\dfrac{A_0}{1 + \dfrac{s}{\omega_0}}}{1 + K\dfrac{A_0}{1 + \dfrac{s}{\omega_0}}}.$$

- Here, A_0 denotes the low-frequency gain and ω_0 the -3 dB bandwidth.

8

Multiplying the numerator and the denominator by $1 + s/\omega_0$ gives

$$\frac{Y}{X}(s) = \frac{A_0}{1 + KA_0 + \dfrac{s}{\omega_0}}$$

$$= \frac{\dfrac{A_0}{1 + KA_0}}{1 + \dfrac{s}{(1 + KA_0)\omega_0}}.$$

We conclude that the closed-loop system now exhibits:

$$\text{Closed-Loop Gain} = \frac{A_0}{1 + KA_0}$$

$$\text{Closed-Loop Bandwidth} = (1 + KA_0)\omega_0.$$

In other words, the gain and bandwidth are scaled by the same factor but in opposite directions, displaying a *constant* product.

9

3. Modification of I/O Impedances

The Problem:

- For ideal signal transfer, an amplifier's input and output impedances should be tailored to the source and load.
- For example, a good voltage amplifier should have a very high input impedance (to avoid loading the source) and a very low output impedance (to drive the load without its gain dropping).
- An open-loop amplifier rarely has these ideal characteristics.

The Solution:

- Negative feedback modifies the input and output impedances by the factor **(1 + Loop Gain)**.
- The direction of change (increase or decrease) depends on the feedback topology.

10

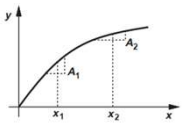
4. Linearity Improvement

The Problem:

- Real amplifiers are not perfectly linear.
- Their gain can change depending on the level of the input signal, as shown in Figure.
- This non-linearity causes signal distortion, which is highly undesirable in applications like audio amplification.

The Solution:

- Negative feedback forces the amplifier to be more linear.
- The closed-loop gain remains much more constant across the input signal range.



(a)

11

The Solution:

- Negative feedback forces the amplifier to be more linear.
- The closed-loop gain remains much more constant across the input signal range.

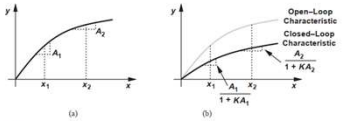


Figure 12.13 (a) Nonlinear open-loop characteristic of an amplifier, (b) improvement in linearity due to feedback.

The closed-loop gain at any point is A / (1 + KA).

- Near x_1, Gain ≈ A_1 / (1 + KA_1) ≈ 1/K
- Near x_2, Gain ≈ A_2 / (1 + KA_2) ≈ 1/K

As long as the loop gain (KA) remains large even when the open-loop gain (A) drops, the closed-loop gain will stay "locked" near the value of 1/K. The variation in the gain is suppressed, resulting in a much more linear input/output relationship, as shown in Figure 12.13(b).

12

The Building Blocks - Four Types of Amplifiers

- We often think of an amplifier as a device that takes a small voltage in and produces a larger voltage out.
- While this is common, it's only one of four possibilities.
- An amplifier is a two-port network that can sense or produce either a **voltage** or a **current**.
- This gives us a 2x2 matrix of amplifier types.

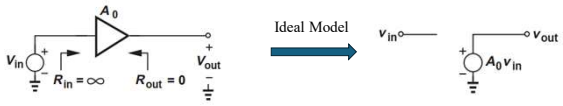
To understand them, we must first understand the concept of ideal impedances.

•**Sensing a Voltage:** To measure a voltage across two points, you use a voltmeter in parallel. To not disturb the original circuit, the voltmeter must draw negligible current. Therefore, an ideal voltage sensor has infinite input impedance (Rin = ∞).
•**Sensing a Current:** To measure a current, you use an ammeter in series with the wire. To not affect the original current, the ammeter must have a negligible voltage drop across it. Therefore, an ideal current sensor has zero input impedance (Rin = 0).
•**Producing a Voltage:** An ideal voltage source (like a perfect battery) produces a constant voltage regardless of the load connected to it. This means it must have zero output impedance (Rout = 0).
•**Producing a Current:** An ideal current source produces a constant current regardless of the load voltage. This requires it to have infinite output impedance (Rout = ∞).

13

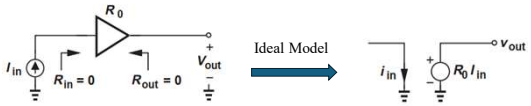
With these rules, let's define the four amplifier types:

1. **Voltage Amplifier (Voltage-Controlled Voltage Source)**
   - **Function:** Senses an input voltage (Vin) and produces a proportional output voltage (Vout).
     - This is the most common type.
   - **Gain Parameter:** Voltage Gain, $A_V = V_{out} / V_{in}$ (Unitless).
   - **Ideal Rin:** ∞ (to sense Vin perfectly).
   - **Ideal Rout:** 0 (to be a perfect voltage source).



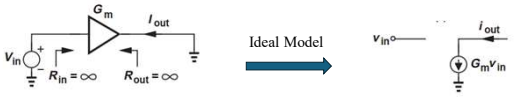
14

2. **Transimpedance Amplifier (Current-Controlled Voltage Source)**
   - **Function:** Senses an input current (Iin) and produces a proportional output voltage (Vout).
     - Useful for converting signals from current sensors (like photodiodes) to voltages.
   - **Gain Parameter:** Transimpedance, $R_0 = V_{out} / I_{in}$ (Units: Ohms, Ω). The name means "transfer impedance."
   - **Ideal Rin:** 0 (to sense Iin perfectly).
   - **Ideal Rout:** 0 (to be a perfect voltage source).



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3. **Transconductance Amplifier (Voltage-Controlled Current Source)**
   - **Function:** Senses an input voltage (Vin) and produces a proportional output current (Iout).
     - A single MOSFET is a good example of a transconductance device.
   - **Gain Parameter:** Transconductance, $G_m = I_{out} / V_{in}$ (Units: Siemens, S or A/V).
   - **Ideal Rin:** ∞ (to sense Vin perfectly).
   - **Ideal Rout:** ∞ (to be a perfect current source).



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4. **Current Amplifier (Current-Controlled Current Source)**
   - **Function:** Senses an input current (Iin) and produces a proportional output current (Iout).
   - **Gain Parameter:** Current Gain, $A_I = I_{out} / I_{in}$ (Unitless).
   - **Ideal Rin:** 0 (to sense Iin perfectly).
   - **Ideal Rout:** ∞ (to be a perfect current source).



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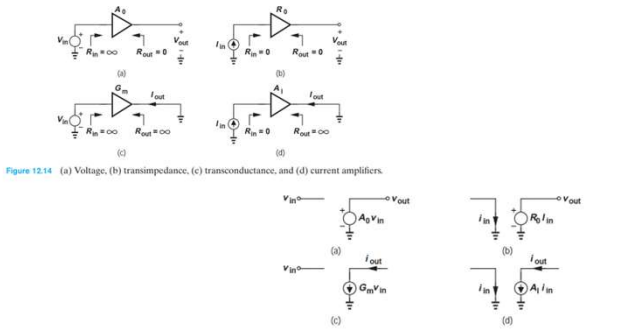


Figure 12.14 (a) Voltage, (b) transimpedance, (c) transconductance, and (d) current amplifiers.

Figure 12.15 Ideal models for (a) voltage, (b) transimpedance, (c) transconductance, and (d) current amplifiers.

18

FEEDBACK TOPOLOGIES

- Now we will use the four amplifier types as the "forward system" (A_1) in a feedback loop.
- A feedback topology is defined by two things:

  1.What quantity is being sensed at the output? (Voltage or Current)
  2.What quantity is being returned to the input? (Voltage or Current)

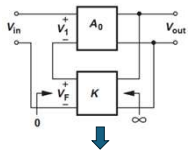
This creates four combinations. The connections are key:

•**Sensing Voltage:** The **feedback** network is connected in **parallel (or shunt)** with the output.
•**Sensing Current:** The **feedback** network is connected in **series** with the output.
•**Returning a Voltage:** The **feedback** signal is subtracted in **series** with the input voltage.
•**Returning a Current:** The **feedback** signal is subtracted in **parallel (or shunt)** with the input current.

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1. Voltage-Voltage Feedback (Series-Shunt)

•**Configuration:** Senses **Voltage** (Shunt at Output), returns **Voltage** (Series at Input).
•**Forward Amplifier:** Requires a **Voltage Amplifier**.



Gain

We first calculate the closed-loop gain. Since

$$V_1 = V_{in} - V_F$$
$$V_{out} = A_0 V_1$$
$$V_F = KV_{out},$$

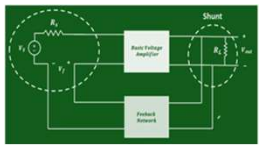
we have

$$V_{out} = A_0(V_{in} - KV_{out}),$$

and hence

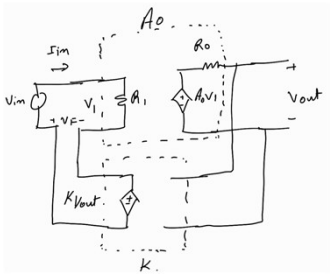
$$\frac{V_{out}}{V_{in}} = \frac{A_0}{1 + KA_0},$$

an expected result.



20

Effective input impedance:



$R_i = \frac{V_{im}}{I_{im}}$ $I_{im} R_{im}$

$V_1 = (V_{im} - V_F)$

$I_{im} R_{im} = V_{im} - K V_{out}$ $\Rightarrow$ $V_1 A_0$ $I_{im} R_{im}$

$\Rightarrow I_{im} R_{im} = V_{im} - K A_0 V_1$

$\Rightarrow I_{im} R_{im} = V_{im} - K A_0 I_{im} R_{im}$

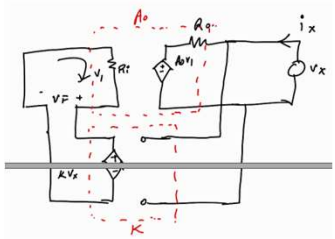
$\Rightarrow I_{im} R_{im} (1 + KA_0) = V_{im}$

$\Rightarrow \boxed{\frac{V_{im}}{I_{im}} = R_{im} (1 + KA_0)}$

Effective input impedance increases by factor of 1+KA0 because of the input series connection at the mixing side

21

Effective output impedance:



$V_F = K V_X$

$V_1 = -V_F$

$V_X = R_0 i_X + A_0 V_1$

$V_X = R_0 i_X - A_0 K V_X$

$V_X (1 + A_0 K) = R_0 i_X$

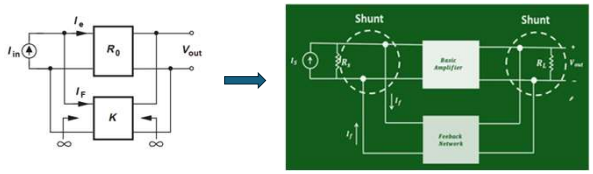
$\Rightarrow \boxed{\frac{V_X}{i_X} = \frac{R_0}{1 + A_0 K}}$

Effective output impedance decreases by factor of 1+A0K because of the input shunt connection at the sensing side

22

2. Voltage-Current Feedback (Shunt-Shunt)

•**Configuration:** Senses **Voltage** (Shunt at Output), returns **Current** (Shunt at Input).
•**Forward Amplifier:** Requires a **Transimpedance Amplifier**.



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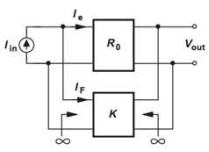
Gain

WKT Ie = Iin - I_F
WKT *Vout = IeRo*
So,

$$V_{out} = (I_{in} - I_F)R_0$$
$$= (I_{in} - KV_{out})R_0,$$

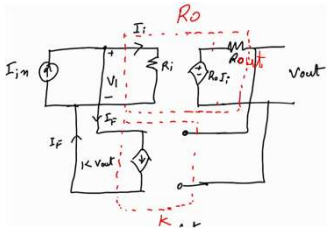
and hence

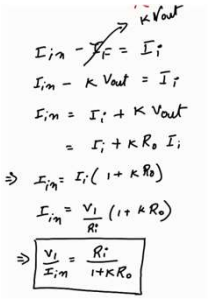
$$\frac{V_{out}}{I_{in}} = \frac{R_0}{1 + KR_0}.$$



24

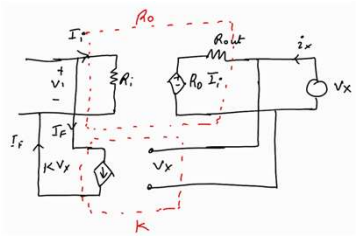
Effective input impedance:

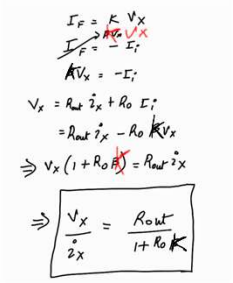




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Effective output impedance:

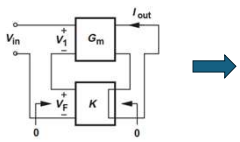


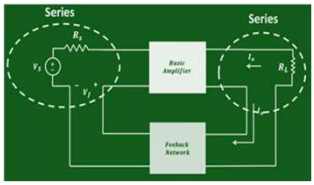


26

3. Current-Voltage Feedback (Series-Series)

•**Configuration:** Senses **Current** (Series at Output), returns **Voltage** (Series at Input).
•**Forward Amplifier:** Requires a **Transconductance Amplifier**.





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Gain

WKT forward system produces a current:
$Iout = Gm(Vin - V_F)$

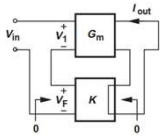
Since VF = KIout, we have

$$I_{out} = G_m(V_{in} - KI_{out})$$

and hence

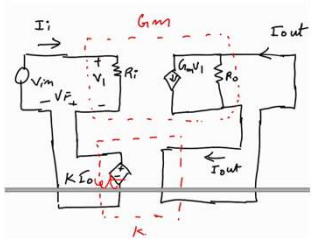
$$\frac{I_{out}}{V_{in}} = \frac{G_m}{1 + KG_m}.$$

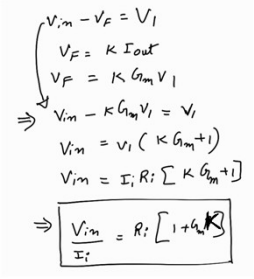
Here $K = V_F/Iout.$



28

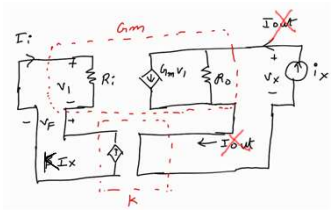
Effective input impedance:

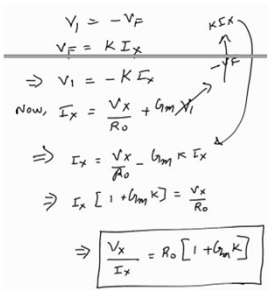




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Effective output impedance:

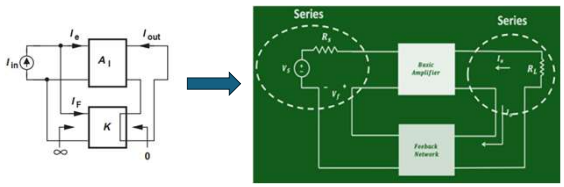




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4. Current-Current Feedback (Shunt-Series)

•**Configuration:** Senses **Current** (Series at Output), returns **Current** (Shunt at Input).
•**Forward Amplifier:** Requires a **Current Amplifier**.



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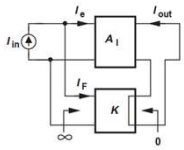
Gain

The current entering the forward amplifier yields:

$$I_{out} = A_I(I_{in} - I_F)$$
$$= A_I(I_{in} - KI_{out})$$

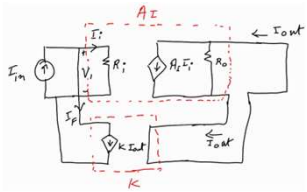
and hence

$$\frac{I_{out}}{I_{in}} = \frac{A_I}{1 + KA_I}. \qquad K = I_F/I_{out}.$$



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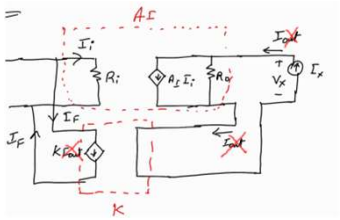
Effective input impedance:



$$I_{in} - I_F = I_i$$
$$I_{in} - K I_{out} = I_i$$
$$I_{in} - K A_I I_i = I_i$$
$$I_{in} = I_i [1 + K A_I]$$
$$I_{in} = \frac{V_i}{R_i}[1 + K A_I]$$
$$\Rightarrow \boxed{\frac{V_i}{I_{in}} = \frac{R_i}{1 + K A_I}}$$

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Effective output impedance:



$$I_F = K I_x$$
$$I_i = -I_F$$
$$I_i = -K I_x$$
Now, $$I_x = A_I I_i + \frac{V_x}{R_o}$$
$$I_x = -A_I K I_x + \frac{V_x}{R_o}$$
$$\Rightarrow \boxed{\frac{V_x}{R_o} = I_x [1 + A_I K]}$$

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